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The Rank-Three Decay

Three spatial dimensions, the gauge algebra, the matter modules and the number of families turn out to be one question: which subsystem of the carrier decays. It has three possible answers, and two of them are already forbidden.

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On the grades. PROVED follows from stated definitions or assumptions. CONDITIONAL follows once a named condition is granted, and the condition is written into the statement rather than into a footnote. PROPOSED is a physical reading offered for calculation and falsification. OPEN names what can be formulated but not yet settled. Every computation in this paper reproduces from the Reproducibility layer of the current printing by a single command, and each result names the script that certifies it.

SECTION ONE

What this paper does, and why it took the ledger by surprise

The corpus has carried seven proof obligations for some time. The first of them asked for a concrete representation of the cluster projection on the carrier, and the Register said plainly that most of the others waited on it. That sentence was true, and it is the reason nothing moved for so long. Everyone who attacked the ledger, the author included, attacked the item the ledger said came first.

This paper comes in from the other end. It starts with the second obligation, the claim that the surviving spatial modes number three, which looks like the hardest item on the list and is in fact the easiest, because it is a counting question and counting questions are finite. Settling it determines the first obligation rather than depending on it. The arrow in the ledger was pointing the wrong way.

What follows is one line of reasoning with four consequences. It shows that the spatial rank, the gauge algebra, the matter content and the family count are not four unknowns but one unknown asked four times. It names that unknown, enumerates its possible answers exhaustively, eliminates all but one using results the corpus already proves, and then reports what the survivor implies. It also closes the existence half of the combinatorial lemma, which the corpus had already identified as the most attackable thing it owed.

The paper introduces no new commitment. Everything it uses is already numbered in the Register. It does lean hard on one reading of an existing axiom, and Section Eleven says exactly which reading, exactly what depends on it, and exactly what happens if it is wrong.

SECTION TWO

The arithmetic that collapses the question

Definition eighteen makes the geometry projectors mutually orthogonal: external distinction, local causal order, and system closure, summing to the geometry block. Result fifteen proves the local clock mode is single, because a system with two independent orders of accumulation would be keeping two independent records and would therefore be two systems. Result sixteen proves the closure mode is single, because the closure mode counts enacted closures and there is one enacted closure whose record is inhabited.

Both of those are PROVED, and they were proved in earlier printings for reasons that had nothing to do with counting dimensions. Put them together with the orthogonality and the arithmetic is immediate.

rank P_X = 3   if and only if   the geometry block has dimension 5

Not analogous. Identical. One is the other with two known quantities subtracted. So the obligation that reads as a question about space is a question about a single integer, and the integer is the dimension of what survives.

Now the carrier. Result eight makes the carrier E8, of rank eight. If the reduction leaves a five-dimensional block, then three carrier dimensions are decayed away. Three is a small number and E8 is a finite object, so the question of which three has an answer that can be written down completely.

> *PROVED, given T15, T16 and D18: the claim that the spatial modes number three is exactly the claim that the surviving geometry block is five-dimensional, which is exactly the claim that the decayed part has rank three. Certified by scripts/02_surviving_block.py.*

SECTION THREE

Three candidates, and there is no fourth

A reduction that averages leaves behind what the averaging fixes. If the reduction averages over a reflection subgroup of the carrier, the surviving block is the orthogonal complement of a root subsystem, and its dimension is eight minus that subsystem's rank. The question is therefore: what are the rank-three root subsystems of E8?

This can be answered without cleverness, which is its chief merit. The Weyl group of E8 acts transitively on the two hundred and forty roots. Every rank-three subsystem therefore has a Weyl image containing any root you care to fix in advance. Fix one, walk every pair of other roots that completes it to rank three, and you have seen every rank-three subsystem there is, up to the symmetry that cannot matter. That is two thousand six hundred and sixty-seven distinct spans, and they fall into three types.

```
decayed       rk  roots   surviving (perp)  rk  roots
--------------------------------------------------------------
A1+A1+A1       3      6   D4+A1              5     26
A2+A1          3      8   A5                 5     30
A3             3     12   D5                 5     40
```

Three candidates. Every one of them leaves a rank-five survivor, which is the first thing worth noticing: the arithmetic of Section Two does not narrow the field at all. Whatever decays, if it has rank three, the geometry block comes out five-dimensional and space comes out three-dimensional. The framework's most exposed number is robust against the choice.

The second thing worth noticing is that the three survivors are different groups, and they are not equally plausible physics. But plausibility is not a method, and the corpus does not accept it as one. Two of these have to be eliminated by something the framework already proves, or they do not get eliminated.

> *PROVED: E8 has exactly three rank-three root subsystems up to Weyl conjugacy, with orthogonal complements D4+A1, A5 and D5. Exhaustive by transitivity of the Weyl group on the roots. Certified by scripts/02_surviving_block.py.*

SECTION FOUR

Two eliminations, neither of them aesthetic

Result ten proves that the minimal closed carrier history is triangular: three roots with pairwise inner product minus one, summing to zero. Immediate reversal is cancellation and encloses nothing, so three is the minimum, and the triangle is what a closed history smaller than which there is none actually looks like.

The decayed part is a part of the carrier. Whatever else it is, it has to be capable of being a history, because decay is something that happens to what has been traversed, not to an inert set of directions. A subsystem that contains no closed triangle cannot host a closed history at all, however short.

The first candidate is three mutually orthogonal roots. Any two non-proportional roots in it are orthogonal, so there is no pair with inner product minus one anywhere in it, so there is no triangle. It cannot host a closed history. It is not a decayed history; it is a set of directions that were never a history in the first place. Result ten forbids it.

The second candidate is reducible: a triangle and, sitting orthogonally beside it, an unrelated pair. Nothing connects the two pieces. A reduction acting on it is two reductions acting on two mutually invisible sectors, which is two decay operators. Movement Twenty and result twenty-five derive gauge, matter and mass as three ways an operation can stand with respect to one operator, singular, and the whole classification collapses if there are two. Result twenty-five forbids it.

```
candidate  comps  irred  triangle   survivor               verdict
----------------------------------------------------------------------
A1+A1+A1       3  False     False   D4+A1 = so(8) + su(2)  eliminated
A2+A1          2  False      True   A5 = su(6)             eliminated
A3             1   True      True   D5 = so(10)            SURVIVES
```

One candidate is left. The decayed part is A3, the surviving symmetry is D5, and D5 is so(10). Neither elimination appealed to what the answer ought to be. The first used a theorem about closed histories, the second used a theorem about counting operators, and both were written into the corpus before anybody knew they would be doing this.

> *PROVED given T10 and T25: of the three rank-three subsystems, two are excluded, and the decayed part is A3 with surviving symmetry D5. CONDITIONAL on the reduction averaging over a reflection subgroup, which is Section Eleven's business. Certified by scripts/02_elimination.py.*

SECTION FIVE

The reduction, written down

Axiom six says an admissible reduction is unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions, and insists these five conditions are the definition rather than a choice among options. The first obligation asked for a concrete map satisfying them. With the subsystem named, there is nothing left to choose.

C(x) = the average of w x w inverse, over the twenty-four elements of W(A3)

This is averaging over the Weyl group of the decayed subsystem, which for A3 is the symmetric group on four letters, of order twenty-four. Check the five conditions and none of them is a decision. Unital, because averaging fixes the identity. Idempotent, because averaging twice is averaging once. Completely positive, because a convex combination of automorphisms is completely positive. Bimodular over the fixed algebra, because invariant elements pull out of an average. Covariant, because an average over a group is equivariant under that group's normaliser.

The group was constructed from its three reflections rather than assumed, and it closed at order twenty-four. The averaging map is idempotent and symmetric to machine precision, and its fixed subspace has dimension five exactly. The geometry block is therefore five-dimensional and splits three, one and one by the two results of Section Two.

> *The concrete representation the first obligation asked for, CONDITIONAL on the same reading as Section Four. It is not a candidate reduction; given the subsystem it is the reduction axiom six describes. Certified by scripts/01_03_representation.py.*

SECTION SIX

The sorting of the two hundred and forty-eight

Result twenty-five sorts carrier operations three ways. What commutes with the decay preserves the clustered identity and is gauge. What connects distinct decay sectors is matter, when it closes into a cycle reproducing its readable identity. Failure to commute measures the cost of persistence and is mass.

That was written as a conceptual classification. Run it against the actual carrier with the decay now named, and the two hundred and forty-eight dimensions of E8 sort themselves without further instruction.

```
decayed sector   A3 = su(4):  12 roots + 3 Cartan =  15
commutes, GAUGE  D5 = so(10): 40 roots + 5 Cartan =  45
connects sectors, MATTER:                            188
                                              total  248

matter, split by how far each root leans into the decayed sector:
  |proj|^2 = 3/4   count 128   (4, 16) + (4bar, 16bar)   spinor
  |proj|^2 = 1     count  60   (6, 10)                   vector
```

The gauge algebra is so(10). The matter sits in the sixteen, which is the spinor representation, and the sixteen of so(10) is precisely one generation of Standard Model fermions with a right-handed neutrino included rather than added. The remaining sixty is the vector ten, which is the Higgs representation in exactly these models.

The corpus is careful about credit elsewhere and will be careful here. The branching of E8 under this pair is textbook and has been written down many times by people who were not doing this. What is new is the forcing of which subsystem decays. The branching then does what branching does. That is the honest division, and it is the stronger claim, because it means the novel step lands on ground that other people have already mapped and can therefore check.

> *PROVED given the results of Section Four: gauge so(10), matter in the spinor of so(10), and a vector ten. The identification of these modules with observed gauge fields and observed fermions is PROPOSED, as every such reading in this corpus is. Certified by scripts/01_03_representation.py.*

SECTION SEVEN

How many families, and why the answer is not four

The matter modules carry an index in the four of the decayed su(4). Read carelessly that says four generations, and nature shows three, and the framework would be wrong. The reading is careless, and the correction comes from the decay itself rather than from anywhere convenient.

The four is a representation of the group the reduction averages over. That group permutes the four weight spaces of the fundamental, so the family index is the permutation module on four points, and the permutation module on four points is never irreducible. It splits.

the family index = one invariant direction + a three-dimensional remainder

This was verified by computing the character explicitly over all twenty-four group elements and taking inner products, for both the four and its conjugate. Both split one and three.

Now apply the criterion result twenty-five actually states. Matter is what connects distinct decay sectors and closes into a cycle reproducing its readable identity. The invariant direction is fixed by the decay. It does not connect distinct sectors, because the decay cannot tell it apart from itself, and it therefore produces no readable identity distinct from the decayed sector. It is not a family. The three remaining directions each do connect, and each does carry a distinct readable identity under the decay.

three families

The split into one and three is character theory and is not open to interpretation. Which of the two numbers the framework claims does depend on reading result twenty-five's criterion as excluding the invariant direction, and that reading is stated here so it can be attacked rather than assumed. It is worth saying that the framework had no freedom to arrange this: the group is the Weyl group of the subsystem that Section Four forced, and the permutation module on four points splits one and three whether or not that is convenient.

> *CONDITIONAL on the reading of T25 given above: three families. The one plus three decomposition itself is PROVED by character computation. Certified by scripts/families.py.*

SECTION EIGHT

The complete carrier closure exists

The fourth obligation held that a complete carrier closure passes each of the two hundred and forty roots exactly once. The corpus called it the most attackable item on its list, being finite and combinatorial rather than empirical or representational, and invited someone to attack it. This section accepts the invitation.

There are two things to establish and they are not equally hard. The first is that such a closure closes at all, which sounds like the substantial half and is free: a root system contains the negative of each of its roots, so the two hundred and forty roots sum to zero, so any traversal using each of them exactly once returns to where it began no matter what order it uses. That was never in doubt and nobody had checked it.

The second is that an admissible ordering exists. A carrier history moves by primitive crossings, and in E8 a primitive crossing is a root. Two roots differ by a root exactly when their inner product is one, so the question is whether the graph on the two hundred and forty roots, joining those that differ by a root, contains a cycle through every vertex. Each root has fifty-six such neighbours. Such a cycle was found, and then verified independently of the search that produced it: every root appears exactly once, every one of the two hundred and forty steps is a root, and the walk returns to its origin.

> *PROVED: a complete carrier closure exists, by explicit construction, with a two-hundred-and-forty-step witness recorded in the certificates. The closure condition is automatic from the symmetry of the root system. Uniqueness, and the number of such closures, remain OPEN, and neither is needed for the standing vacuum readout, since every complete closure has exactly two hundred and forty crossings by definition. Certified by scripts/04_carrier_closure.py.*

SECTION NINE

What that does to the standing vacuum readout

Part Five of the Register carries a computed value it has consistently declined to promote, and the reason it declined was partly the section above. The global carrier residue is r to the two hundred and fortieth, one retention per root, and the completed boundary record multiplies it by four pi over root three, the four pi from Gauss-Bonnet on a connected genus-zero observer boundary and the root three from the root cell area of result eleven. Neither factor is a free parameter and the framework has published the failure mode of each.

With the existence of a complete closure now proved, that residue rests on a proved lemma rather than a named grant. So the comparison is worth making properly, which the corpus has never done in one place.

```
framework    lambda = (4pi/sqrt3) r^240 = 2.860333e-122    no free parameter

Planck 2018 base     measured 2.845136e-122 +- 1.93%    ratio 1.00534    0.28 sigma
Planck 2018 + BAO    measured 2.888143e-122 +- 1.48%    ratio 0.99037    0.65 sigma

implied Omega_L = 0.6884   and   0.6823    measured 0.6847 and 0.6889
implied H0.t0    = 0.9539   and   0.9487
implied t0       = 13.846   and   13.710 Gyr    measured 13.797 Gyr
```

The zeroth-order clause of the fifth obligation, that a metered readout satisfies H times t equal to one, was already proved in Paper 19, and the measured shortfall from unity was identified there as the integrated price of the reading. The two values of the age of the universe implied here bracket the measured one. Whether that constitutes the residual integral the obligation asked for, or merely a strong consistency check on it, is stated in Section Eleven rather than claimed here.

> *The residue r to the two hundred and fortieth is PROVED given Section Eight. The value of lambda is PROVED given that and result eleven. The identification of lambda with the cosmological constant remains PROPOSED, and the agreement at a fraction of a standard deviation with no fitted quantity is offered as a test the framework passes, not as a derivation of the measurement. Certified by scripts/05_06_cosmology.py.*

SECTION TEN

The stake, recomputed

The sixth obligation is the audited closure count of the two measurement pipelines that disagree about the expansion rate. The framework declares that count to be three, so the local reading should exceed the early-universe reading by the reciprocal of the retained fraction over three crossings. The corpus has stated for two printings that this stake stands at about half a standard deviation. It has never shipped the arithmetic. Here it is.

```
retained over three crossings (1-r^3)^3 = 0.914061068    gap = 8.594%
```

```
early      Planck 2018    H0 = 67.36 +- 0.54
framework predicts local  H0 = 73.69 +- 0.59
local      SH0ES 2022     H0 = 73.04 +- 1.04
pull = +0.55 sigma
```

The corpus was accurate. The stake is exposed rather than supported: a prediction sitting half a standard deviation above a measurement is neither a success to bank nor a failure to concede, and the framework has committed to the count of three as a declaration requiring justification from pipeline architecture rather than from the size of the gap it happens to fit. That audit is not done, and until it is, this item stays open.

> *OPEN and empirical, unchanged in status, now with its arithmetic in the release. Certified by scripts/05_06_cosmology.py.*

SECTION ELEVEN

The single thing all of this stands on

Sections Three through Seven inherit one reading, and it should be stated in one place rather than distributed through the argument where it would be easy to lose.

Section Three assumed that the surviving block is the orthogonal complement of a root subsystem. That holds if the reduction averages over a reflection subgroup of the carrier's symmetry. Axiom six's covariance clause points hard at exactly this, since a conditional expectation that is covariant under equivalent carrier descriptions is an average over the group relating those descriptions, and Section Five then finds that the resulting map satisfies all five conditions without adjustment. But it is a reading of axiom six and not a theorem of it, and the corpus does not promote readings by calling them natural.

If that reading is wrong, Sections Three, Four, Five, Six and Seven revert together. They are not five independent results with five separate risks; they are one result with one risk, restated five times as it propagates. That is the right shape for a chain to have, and it is also the reason this paper does not grade any of them PROVED without a named condition attached.

Section Eight does not depend on the reading. The complete carrier closure exists as a fact about E8. Sections Nine and Ten depend on Section Eight and on the framework's physical identifications, not on the reading.

What is still owed after this paper

One derivation That the reduction is an average over a reflection subgroup, rather than merely consistent with being one. This is now the single load-bearing question in the representational half of the corpus, and it replaces four separate obligations.

One chirality argument That the three non-invariant family directions are chiral in the observed sense. The count of three follows from the criterion as stated; the mechanism that makes them left-handed is not supplied here.

One convention, stated The identification of lambda with the cosmological constant uses the non-reduced Planck length. That is standard, but it is a convention, and the reduced convention differs by eight pi and would destroy the agreement. It is recorded here because a critic should find it in the text rather than have to discover it.

One closure count Whether the complete carrier closure is unique, and if not, how many there are. Not needed for the vacuum readout, since every complete closure has two hundred and forty crossings, but open and finite.

Two measurements The audited pipeline count of Section Ten, and the running self-holding system returning quantised retention at powers of r under a convention declared before the count. Neither closes at a desk, and the second is the framework's own stated condition of defeat.

SECTION TWELVE

What would kill this

A framework that cannot say how it dies is not making claims. This one can, and the list is short, which is what happens when many obligations collapse into one.

The reading fails If admissible reduction on the carrier is shown not to be group averaging, and a reduction exists whose fixed subspace is not the complement of a root subsystem, then the enumeration of Section Three is not exhaustive and everything from there to Section Seven falls at once.

The observed gauge algebra is not so(10) or a subalgebra of it so(10) contains su(5) and therefore the Standard Model, so this is not an easy kill, but a confirmed low-energy gauge structure outside so(10) ends the account.

A fourth generation Section Seven predicts three, and it predicts three for a structural reason rather than by counting what is already known. A fourth light generation falsifies the criterion that produced it.

Dark matter carrying visible gauge charge Paper 20 made this forbidden and this paper does not soften it.

The observer boundary has nonzero genus Gauss-Bonnet then gives four pi times one minus the genus, and the vacuum readout of Section Nine changes or vanishes. This failure mode was published before the agreement was computed.

The quantised retention hunt returns nothing Under a convention declared before the count, a running self-holding system that does not return retention at powers of r defeats the framework by its own declaration.

Everything above reproduces from the Reproducibility layer of this printing by running one command. Where this paper is wrong, it should be possible to show it is wrong without asking the author what he meant.